

**Computer Networks**

**Program: BSCS**

**Topic**

**Routing Protocols**

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# Routing

- Routing is **the process of selecting a path for traffic in a network or between or across multiple networks.**
- Broadly, routing is performed in many types of networks, including circuit-switched networks, such as the public switched telephone network (PSTN), and computer networks, such as the Internet.

# Static Routing

Static routing is a process in which we have to manually add routes to the routing table.

# Default Routing

- This is the method where the router is configured to send all packets towards a single router (next hop). It doesn't matter to which network the packet belongs, it is forwarded out to the router which is configured for default routing.
- It is generally used with stub routers.
- A stub router is a router that has only one route to reach all other networks.

# Dynamic Routing

- Dynamic routing makes automatic adjustments of the routes according to the current state of the route in the routing table.
- Dynamic routing uses protocols to discover network destinations and the routes to reach them.
- [RIP](#) and [OSPF](#) are the best examples of dynamic routing protocols.
- Automatic adjustments will be made to reach the network destination if one route goes down.

# Features of Dynamic Routing Protocol

- The routers should have the same dynamic protocol running in order to exchange routes.
- When a router finds a change in the topology then the router advertises it to all other routers.

# Advantages of Dynamic Routing

- Easy to configure.
- More effective at selecting the best route to a destination remote network and also for discovering remote network.

# Disadvantages of Dynamic Routing

- Consumes more bandwidth for communicating with other neighbors.
- Less secure than static routing.



# Distance-Vector Routing Protocol

- A distance-vector routing protocol in data networks determines the best route for data packets based on distance.
- Distance-vector routing protocols **measure the distance by the number of routers a packet has to pass**, one router counts as one hop.
- Examples of Distance-Vector Routing Protocols are RIP, EIGRP

# Routing Protocol Vs Routed Protocols

A Routed Protocol is a **protocol by which data can be routed.**

Where as

A Routing Protocol, on the other hand, is only used between routers. Its purpose is to help routers building and maintain routing tables

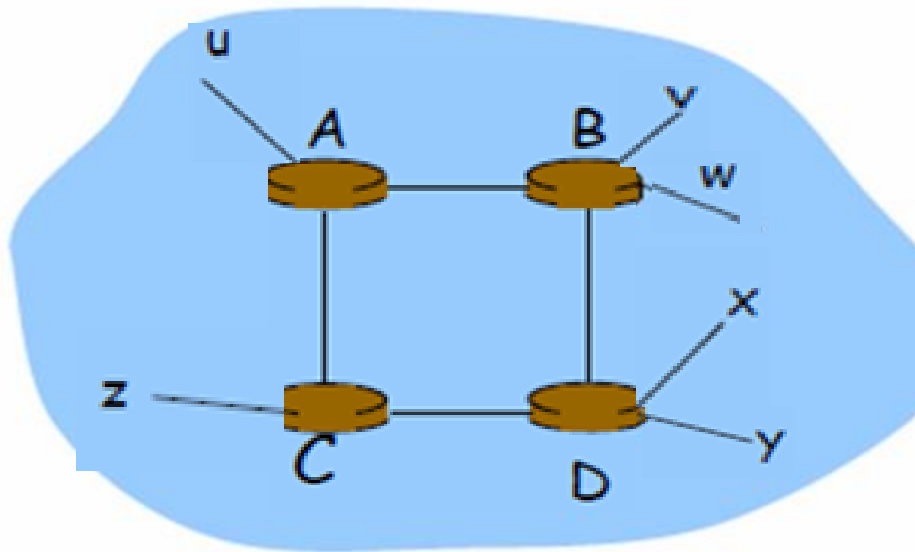
# Convergence

- Convergence is the state of a set of routers that have the same topological information about the internetwork in which they operate.
- Convergence is **part of the routing table update process.**
- When a link fails or changes, updates are sent across the network that describe changes in the network topology. Each router then runs a routing algorithm to recompute routes and build new routing tables based on this information.

# **Routing Information Protocol (RIP)**

# Routing Information Protocol

- The Routing Information Protocol (RIP) is an intra-domain (interior) routing protocol used inside an autonomous system.
- It is a very simple protocol based on distance vector routing.
- RIP implements distance vector routing directly with some considerations.



| <u>Router A</u> |              |             |
|-----------------|--------------|-------------|
| <u>Dest.</u>    | <u>Hops.</u> | <u>Next</u> |
| u               | 1            | -           |
| v               | 2            | B           |
| w               | 2            | B           |
| x               | 3            | C           |
| y               | 3            | C           |
| z               | 2            | C           |

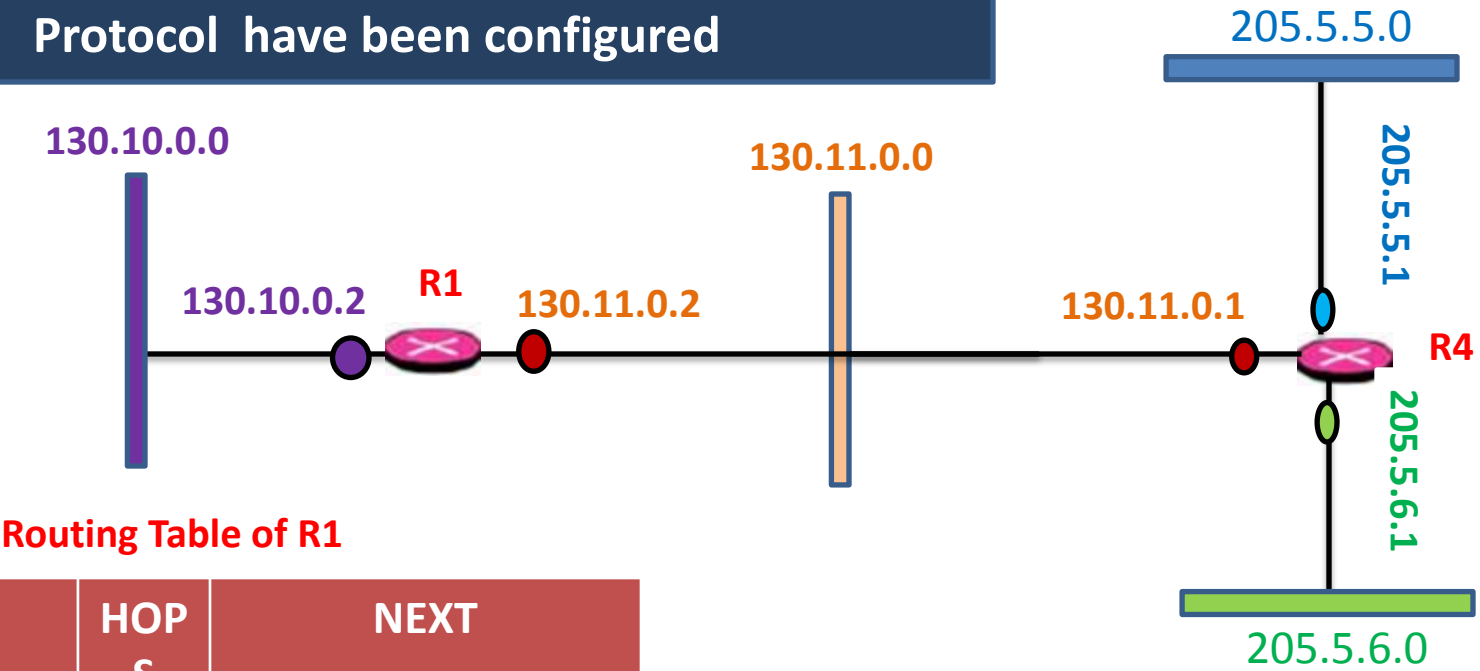
# Routing Information Protocol

- The **destination** in a routing table is a network, which means the first column defines a **network address**
- The metric used is **distance (hop count)** which is defined as the number of links (networks) to reach the destination.
  - **Infinity is defined as 16**, which means that any route in an autonomous system using RIP cannot have more than 15 hops.
- The **next-node** column defines the address of the router to which the packet is to be sent to reach its destination

# RIP Operation

- Router sends update message to neighbors every 30 sec
- A router expects to receive an update message from each of its neighbors within 180 seconds in the worst case [**then route expires**]
- If router does not receive update message from neighbor X within this limit,
  - it assumes the link to X has failed and sets the corresponding minimum cost to 16 (infinity)
- Based on convergence speeded up by triggered updates
  - neighbors notified immediately of changes in distance vector table

Topology And Routing Tables when there is no Protocol have been configured



Routing Table of R1

| DEST.      | HOP S | NEXT  |
|------------|-------|-------|
| 130.10.0.0 | 1     | - - - |
| 130.11.0.0 | 1     | - - - |

Routing Table of R4

| DEST.      | HOP S | NEXT  |
|------------|-------|-------|
| 205.5.5.0  | 1     | - - - |
| 205.5.6.0  | 1     | - - - |
| 130.11.0.0 | 1     | - - - |



# Topology And Routing Tables when RIP Protocol has been configured

130.10.0.0

130.11.0.0

205.5.5.0

130.10.0.2

R1

130.11.0.2

130.11.0.1

205.5.5.1

R4

205.5.6.1

205.5.6.0

Routing Table of R1

When Router R4 Sends  
an Update of its Table  
to Router R1

Routing Table of R4

DEST.

HOP  
S

NEXT

130.10.0.0

1

---

130.11.0.0

1

---

DEST.

HOP  
S

NEXT

205.5.5.0

1

---

205.5.6.0

1

---

130.11.0.0

1

---

# Topology And Routing Tables when RIP Protocol has been configured

130.10.0.0

130.10.0.2

R1

130.11.0.2

130.11.0.0

130.11.0.1

205.5.5.0

205.5.5.1

R4

205.5.6.1

205.5.6.0

Routing Table of R1

| DEST.      | HOP S | NEXT       |
|------------|-------|------------|
| 130.10.0.0 | 1     | ---        |
| 130.11.0.0 | 1     | ---        |
| 205.5.5.0  | 2     | 130.11.0.1 |
| 205.5.6.0  | 2     | 130.11.0.1 |

When Router R4 Sends an Update of its Table to Router R1

Routing Table of R1

| DEST.      | HOP S | NEXT |
|------------|-------|------|
| 205.5.5.0  | 1     | ---  |
| 205.5.6.0  | 1     | ---  |
| 130.11.0.0 | 1     | ---  |

# Topology And Routing Tables when RIP Protocol has been configured

130.10.0.0

130.10.0.2

R1

130.11.0.2

130.11.0.0

130.11.0.1

205.5.5.1

R4

205.5.6.1

205.5.5.0

205.5.6.0

Routing Table of R1

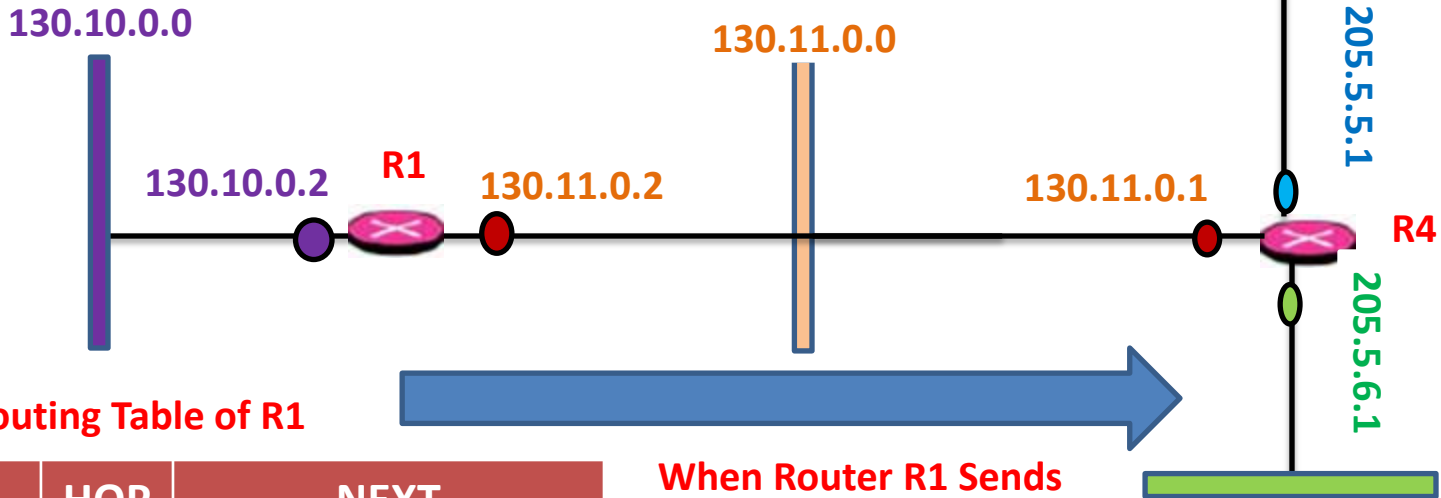
| DEST.      | HOP S | NEXT       |
|------------|-------|------------|
| 130.10.0.0 | 1     | ---        |
| 130.11.0.0 | 1     | ---        |
| 205.5.5.0  | 2     | 130.11.0.1 |
| 205.5.6.0  | 2     | 130.11.0.1 |

When Router R1 Sends an Update of its Table to Router R4

Routing Table of R1

| DEST.      | HOP S | NEXT |
|------------|-------|------|
| 205.5.5.0  | 1     | ---  |
| 205.5.6.0  | 1     | ---  |
| 130.11.0.0 | 1     | ---  |

# Topology And Routing Tables when RIP Protocol has been configured



Routing Table of R1

| DEST.      | HOP S | NEXT       |
|------------|-------|------------|
| 130.10.0.0 | 1     | ---        |
| 130.11.0.0 | 1     | ---        |
| 205.5.5.0  | 2     | 130.11.0.1 |
| 205.5.6.0  | 2     | 130.11.0.1 |

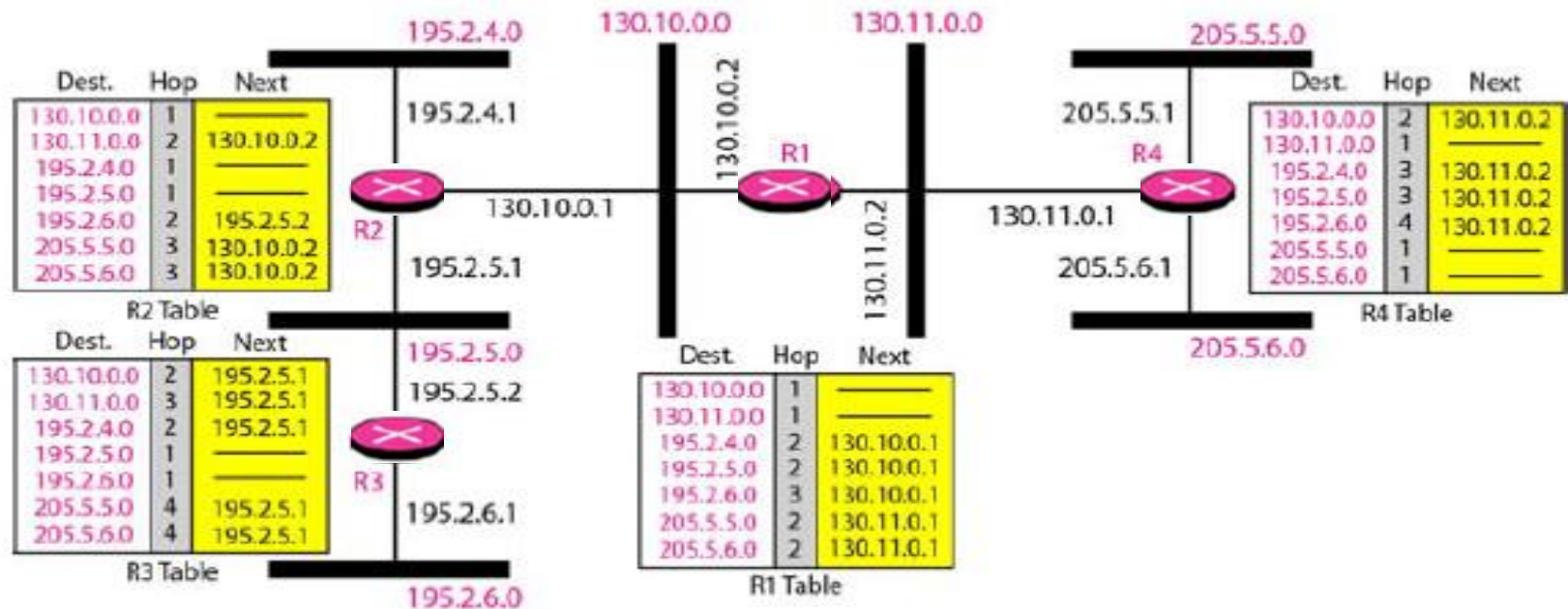
When Router R1 Sends an Update of its Table to Router R4

Routing Table of R1

| DEST.      | HOP S | NEXT       |
|------------|-------|------------|
| 205.5.5.0  | 1     | ---        |
| 205.5.6.0  | 1     | ---        |
| 130.11.0.0 | 1     | ---        |
| 130.10.0.0 | 2     | 130.11.0.2 |

Routing Tables have been converged

# Example of a domain using RIP



- Autonomous system with seven networks and four routers with table of each router.
- Routing table for R1 has seven entries to show how to reach each network in the autonomous system.
- Router is directly connected to networks 130.10.0.0 and 130.11.0.0 which means that there are no next-hop entries for these two networks.

# RIP Updating Algorithm

**Receive a response RIP message**

1. Add one hop to the hop count for each advertised destination.
2. Repeat the following steps for each advertised destination:

**If (destination is not in the routing table)**

        Add the advertised information to the table.

**Else**

**If (next-hop field is the same)**

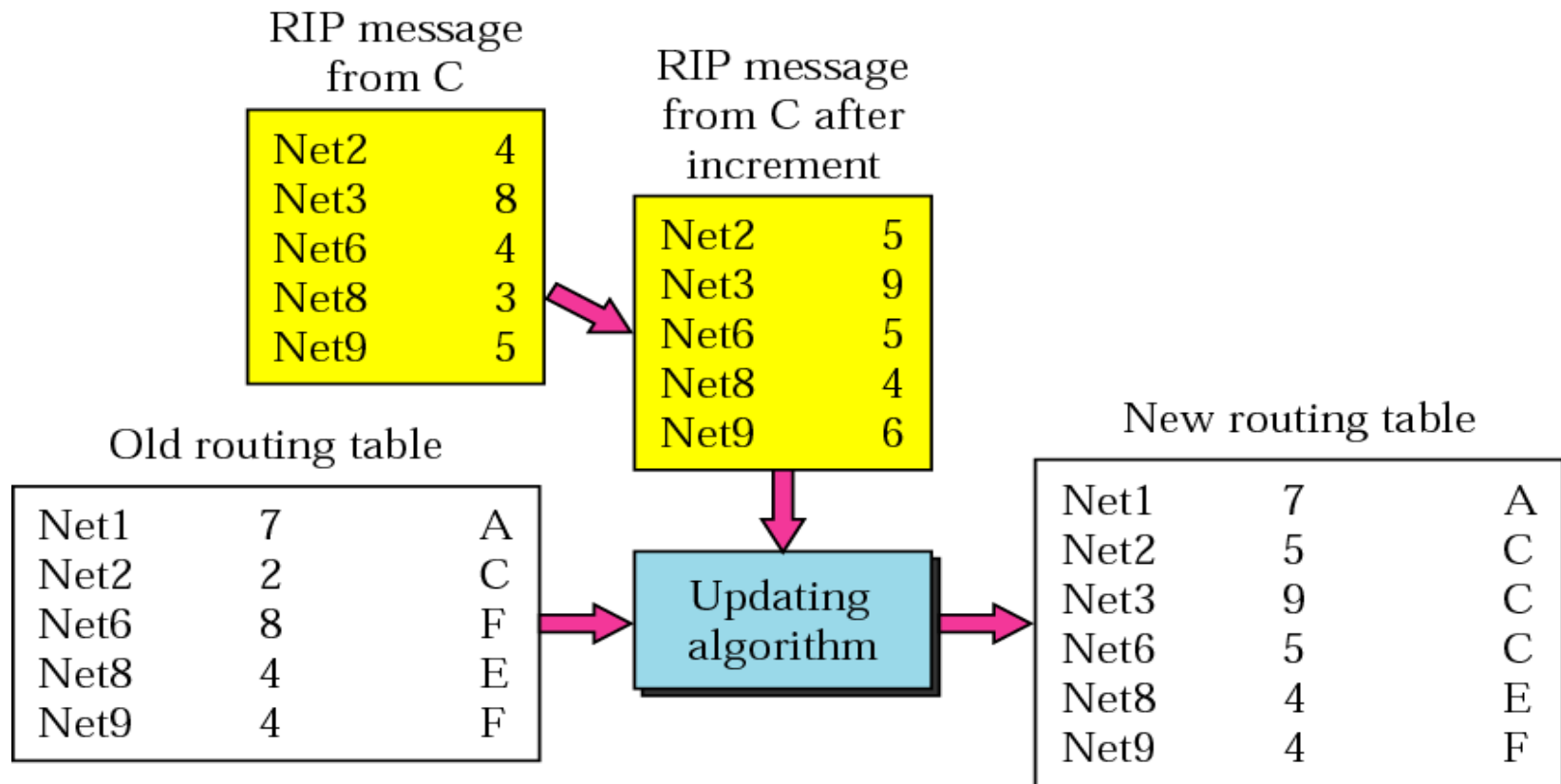
            Replace entry in the table with the new one.

**Else**

**If (new hop count < the one in the table)**

                Replace entry in the routing table.

# Example of updating a routing table



Net1: No news, do not change

Net2: Same next hop, replace

Net3: A new router, add

Net6: Different next hop, new hop count smaller, replace

Net8: Different next hop, new hop count the same, do not change

Net9: Different next hop, new hop count larger, do not change

# Example of updating a routing table

RIP Message from  
Router C to D

|      |   |
|------|---|
| Net2 | 4 |
| Net3 | 8 |
| Net6 | 4 |
| Net8 | 3 |
| Net9 | 5 |

RIP Message from  
Router C After  
Increment

|      |   |
|------|---|
| Net2 | 5 |
| Net3 | 9 |
| Net6 | 5 |
| Net8 | 4 |
| Net9 | 6 |

Old Routing  
Table of Router D

|      |   |   |
|------|---|---|
| Net1 | 7 | A |
| Net2 | 2 | C |
| Net6 | 8 | F |
| Net8 | 4 | E |
| Net9 | 4 | F |

Updating  
Algorithm

New Routing Table of  
Router D

|      |   |   |
|------|---|---|
| Net1 | 7 | A |
| Net2 | 5 | C |
| Net3 | 9 | C |
| Net6 | 5 | C |
| Net8 | 4 | E |
| Net9 | 4 | F |



# Example of updating a routing table

RIP Message from  
Router C to D

|      |   |
|------|---|
| Net2 | 4 |
| Net3 | 8 |
| Net6 | 4 |
| Net8 | 3 |
| Net9 | 5 |

RIP Message from  
Router C After  
Increment

|      |   |
|------|---|
| Net2 | 5 |
| Net3 | 9 |
| Net6 | 5 |
| Net8 | 4 |
| Net9 | 6 |

Old Routing  
Table of Router D

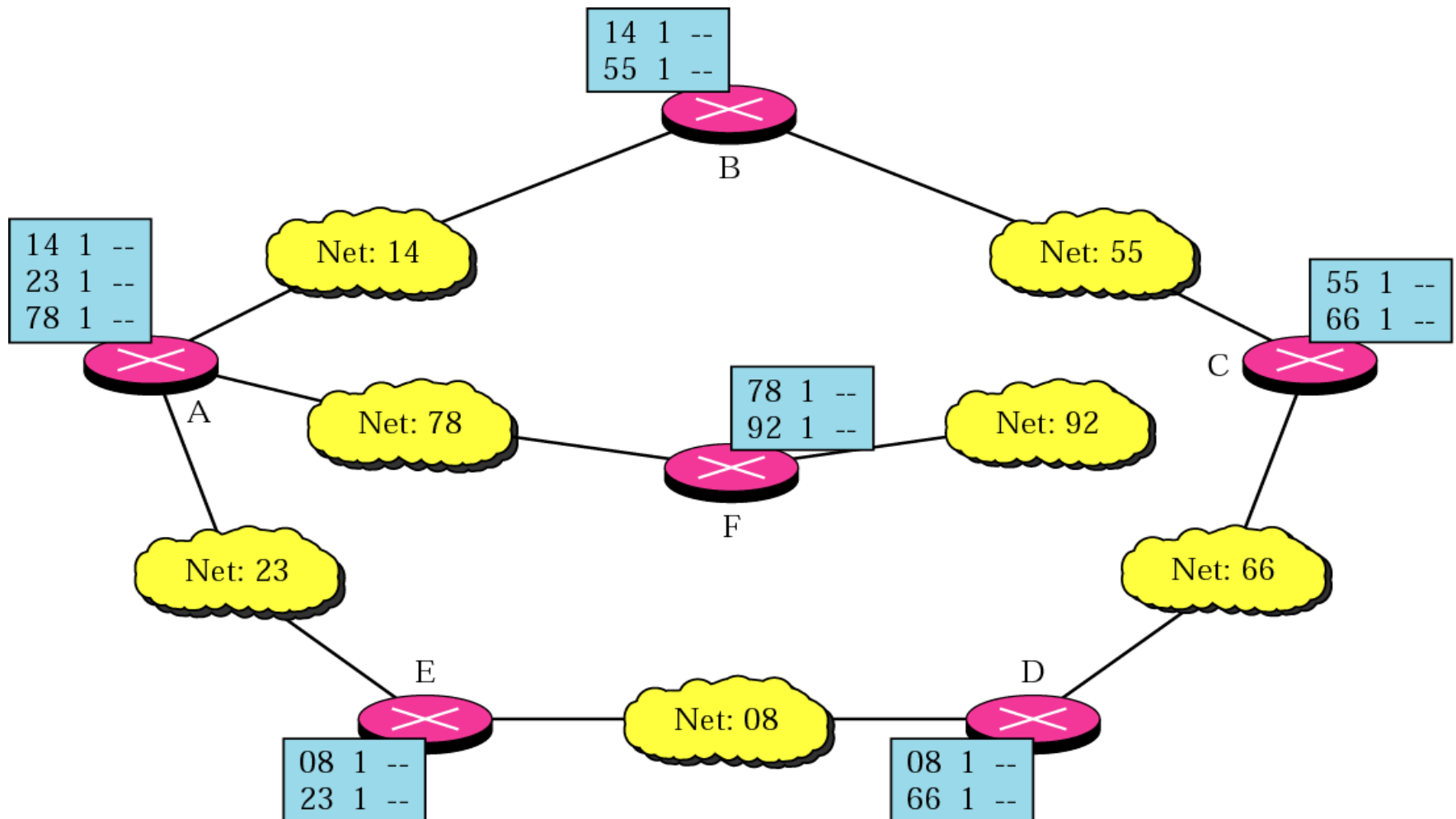
|      |   |   |
|------|---|---|
| Net1 | 7 | A |
| Net2 | 2 | C |
| Net6 | 8 | F |
| Net8 | 4 | E |
| Net9 | 4 | F |

New Routing Table of  
Router D

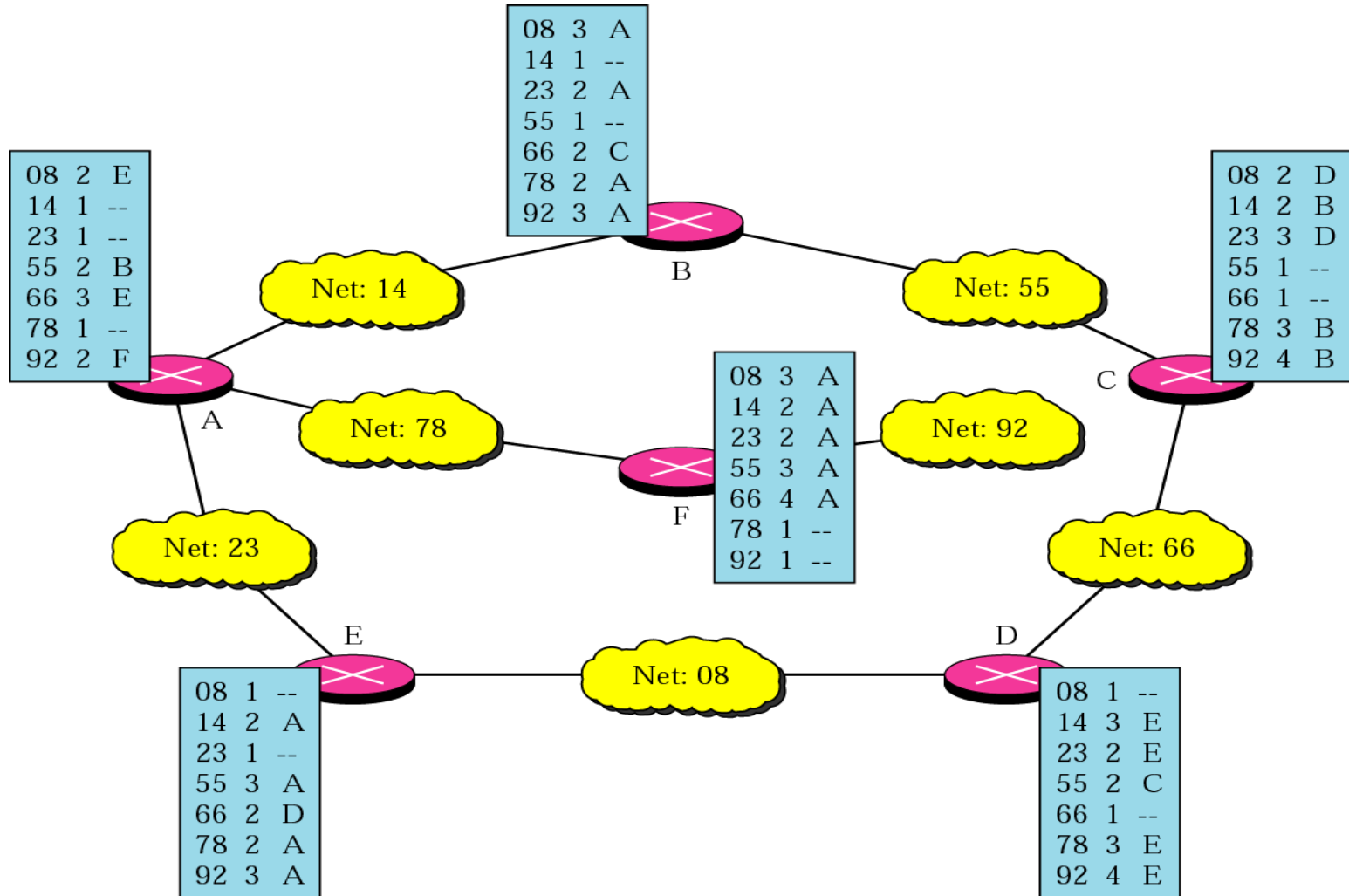
|      |   |   |
|------|---|---|
| Net1 | 7 | A |
| Net2 | 5 | C |
| Net3 | 9 | C |
| Net6 | 5 | C |
| Net8 | 4 | E |
| Net9 | 4 | F |

Updating  
Algorithm

# Initial routing tables in a small autonomous system



# Final routing tables (previous figure)



- RIP is implemented as a process that uses the services of UDP on the well known port 520
- It runs in the background (a daemon process)
- RIP routing protocol helps IP in routing datagrams through AS
  - So RIP messages are encapsulated inside UDP user datagrams; which in turn are encapsulated inside IP datagrams
- Two versions are RIP-1 and RIP-2
  - RIP-2 is backward compatible with RIP-1